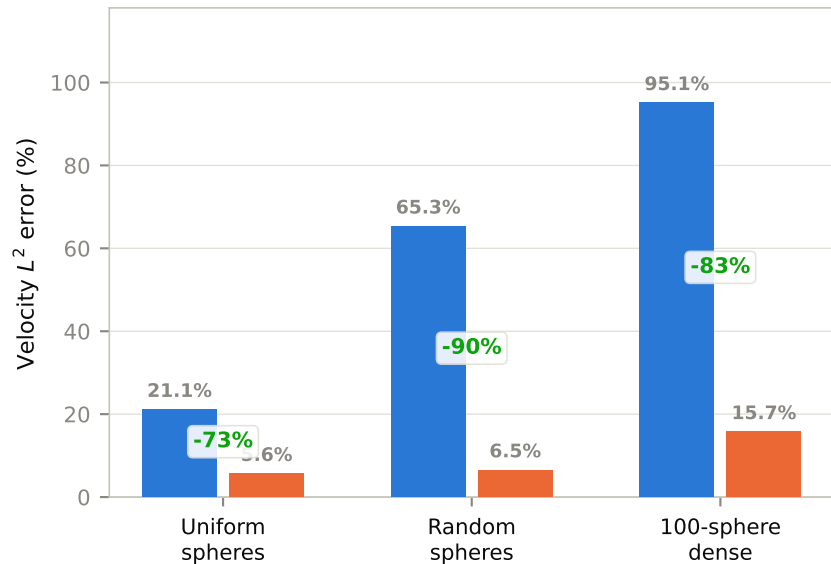
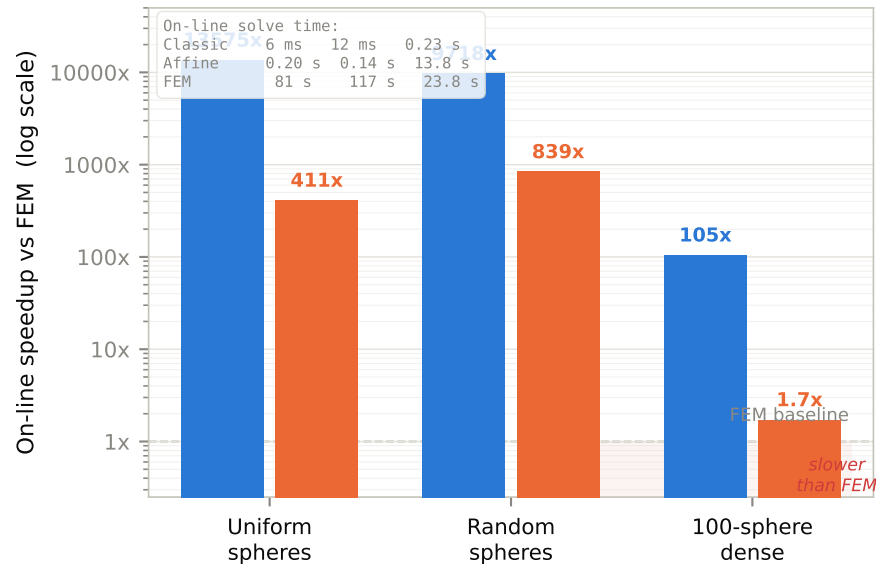


(a) Velocity L^2 error

Uniform: 12,203 cells, 144 ifaces | Random: 15,249 cells, 114 ifaces | 100-sphere: 45,191 cells, 537 ifaces

(b) On-line speedup vs monolithic FEM

On-line solve time:
Classic 6 ms 12 ms 0.23 s
Affine 0.20 s 0.14 s 13.8 s
FEM 81 s 117 s 23.8 s

Same mesh for all methods. Error: 6th-order quadrature, per-element, no interface averaging. On-line speedup = $t_{FEM} / t_{on-line}$ (Schur assembly + solve + reconstruction; offline library pre-built). Green labels: L^2 reduction Classic $\rightarrow$ Affine.